PetCloud

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SPECIAL TOPICS IN COMPUTING · TRENDING INFORMATION TECHNOLOGIES

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INTRODUCTION

PetCloud in a Nutshell

A Centralized Adoption Hub

Connects prospective pet parents directly with animal shelters, rescuers, and independent caretakers.

- **Two User Segments:** Tailored portals for adopters and organizations.
- **Cloud-Native:** Hosted entirely on scalable AWS services.
- **Focused Scope:** Core workflow optimization over feature bloat.



Business Problem

The Challenge: A Fragmented Adoption Process

For Potential Adopters

Scattered Media: Disconnected social channels and listings.

Lost Status: Opaque application review progress.

Outdated Profiles: Unclear animal availability and health history.

For Shelters & Rescuers

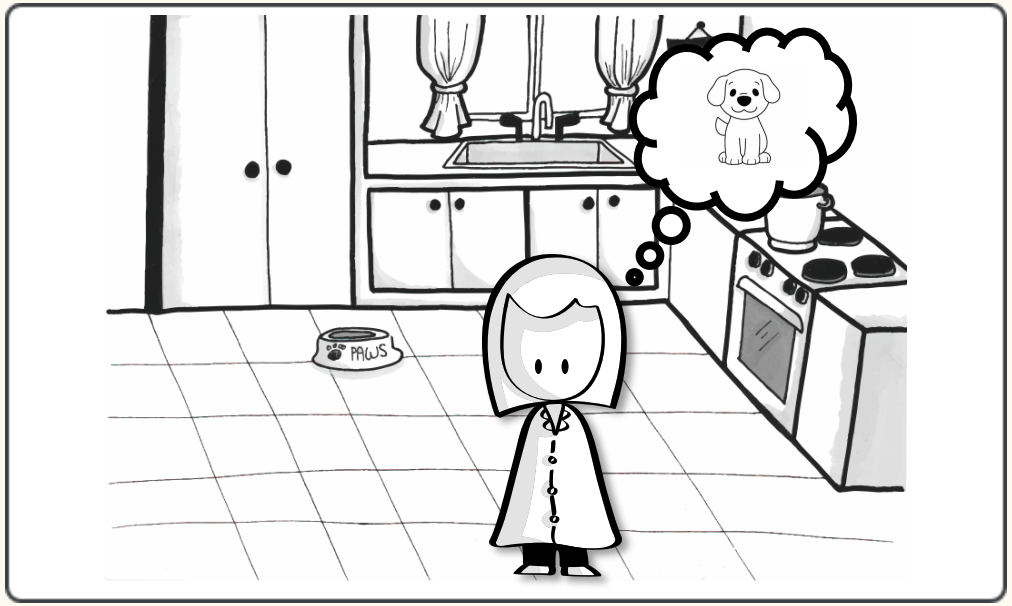
Manual Records: Heavy reliance on spreadsheets and notes.

Informal Inboxes: Inquiries buried in multiple message apps.

Tracking Overhead: Difficult monitoring of adoption lifecycles.

Business Problem

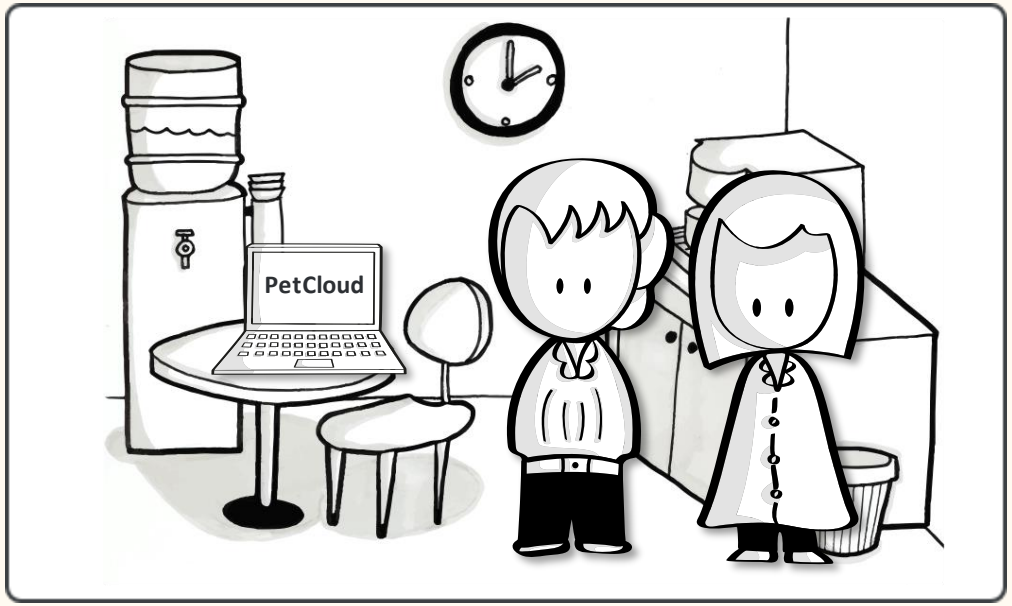
Storyboard: Problem & Discovery



SCENE 01

The Frustrating Search

Marina dreams of adopting a dog, but her daily life is consumed by the exhaustion of trying to find information. She browses through informal channels and dozens of outdated social networks. The process feels distant, cold, and highly fragmented.

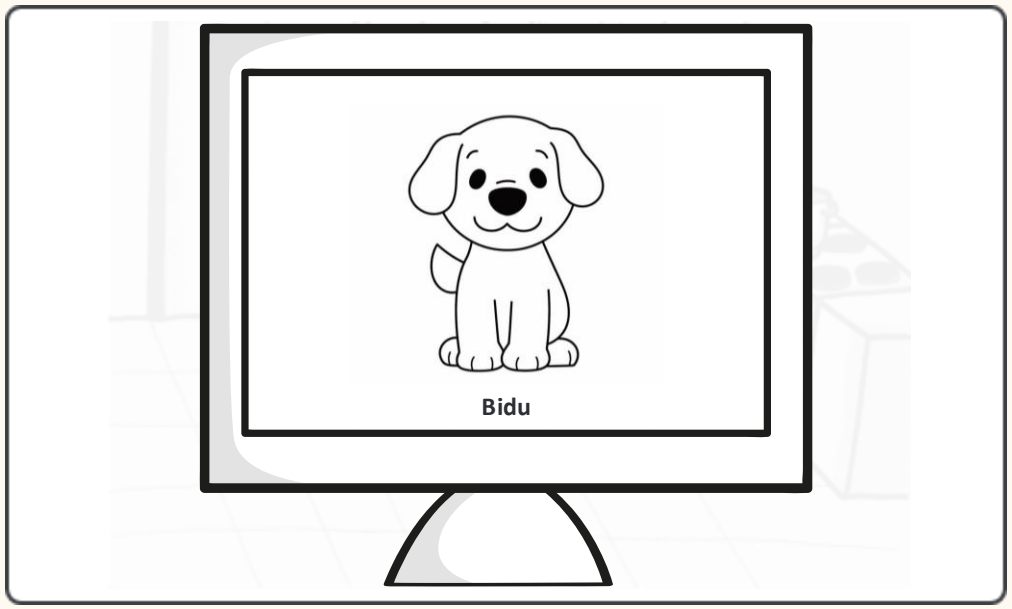


SCENE 02

A Light at the End of the Tunnel

A colleague notices her frustration and sends the PetCloud link. Marina accesses the platform and immediately feels relieved: the environment is welcoming, centralized, and easy to use.

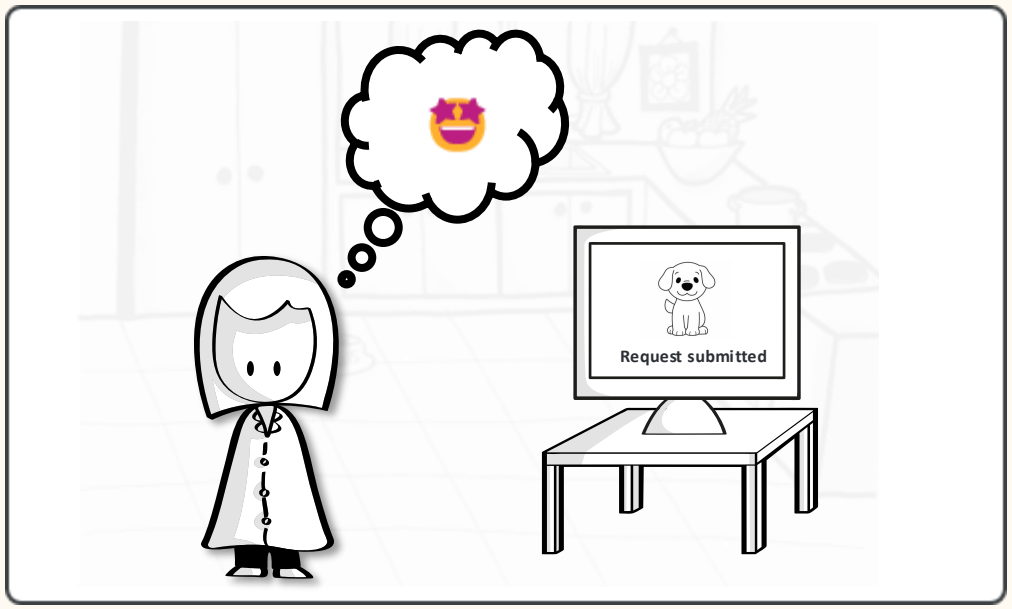
Storyboard: Experience & Action



SCENE 03

The Perfect Match

She uses the platform's filters (age, size, and species). Quickly, she sees Bidu's photos, flawlessly loaded. What was once an exhausting search is now a guided journey full of hope.



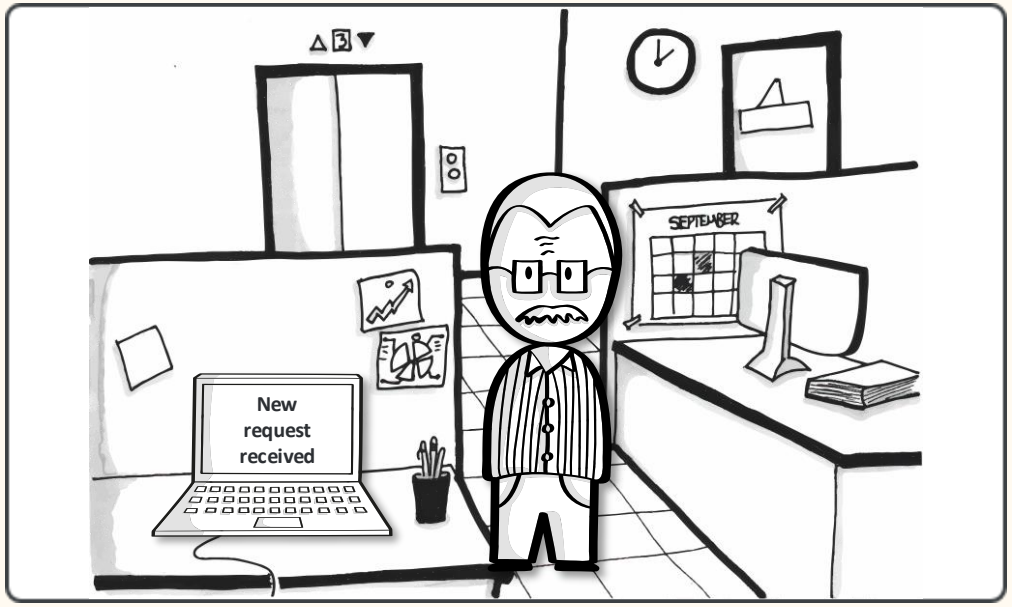
SCENE 04

The Human-Centered Request

Without needing to fill out loose forms or send messages in the dark, Marina submits her adoption request directly through the platform. She feels safe and heard.

Business Problem

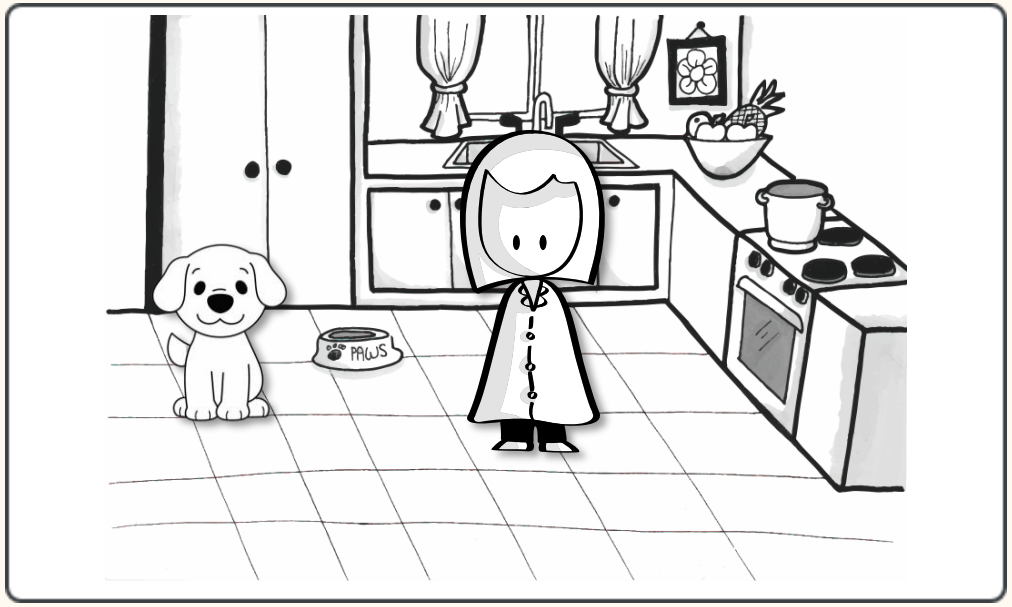
Storyboard: Organization & Happy Ending



SCENE 05

Organization that Saves Lives

At the NGO, Carlos receives Marina's request in a structured way. Instead of wasting hours reconciling manual spreadsheets and lost emails, he approves the profile with a few clicks, having more free time to care for the rescued animals.

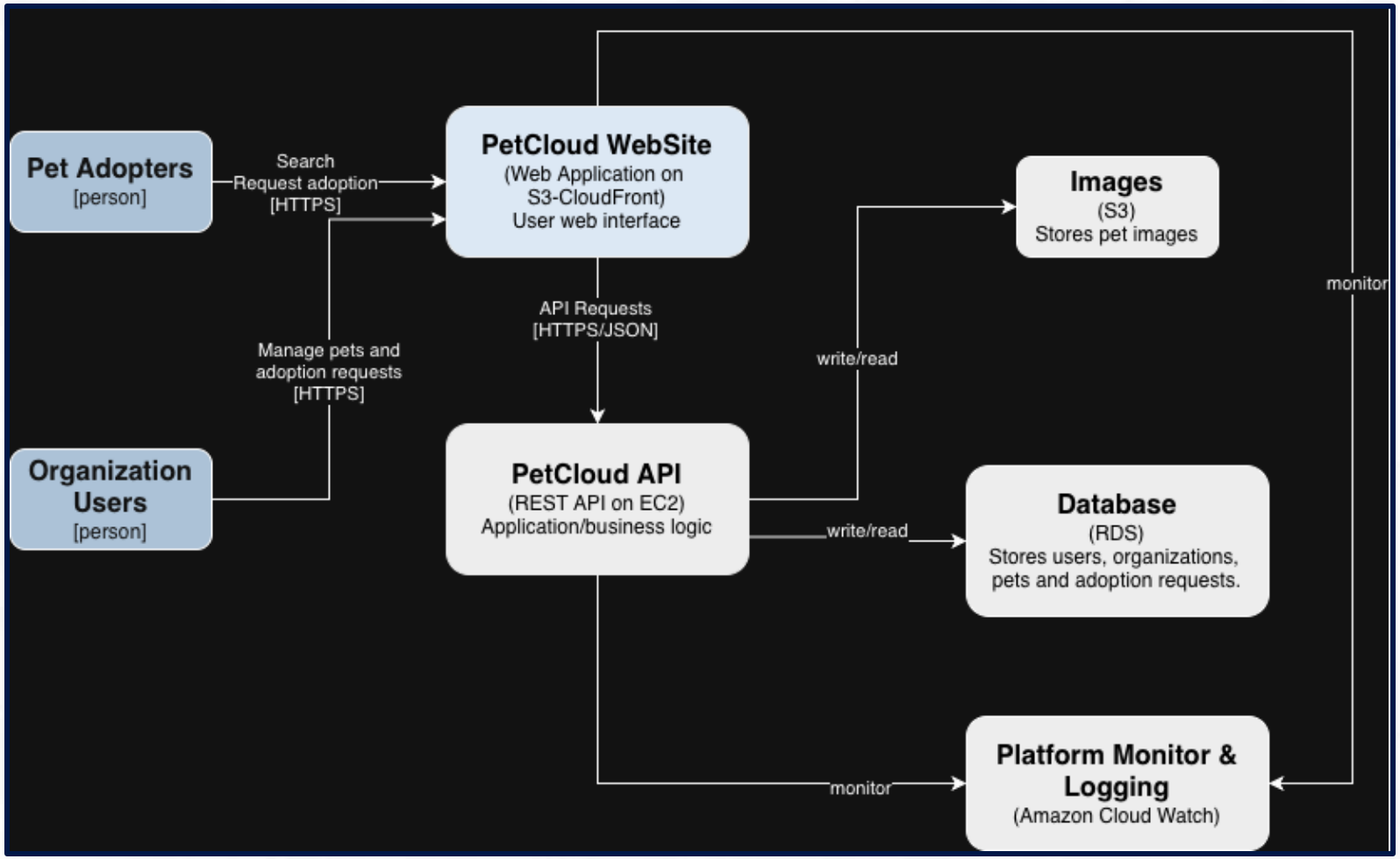


SCENE 06

A New Beginning

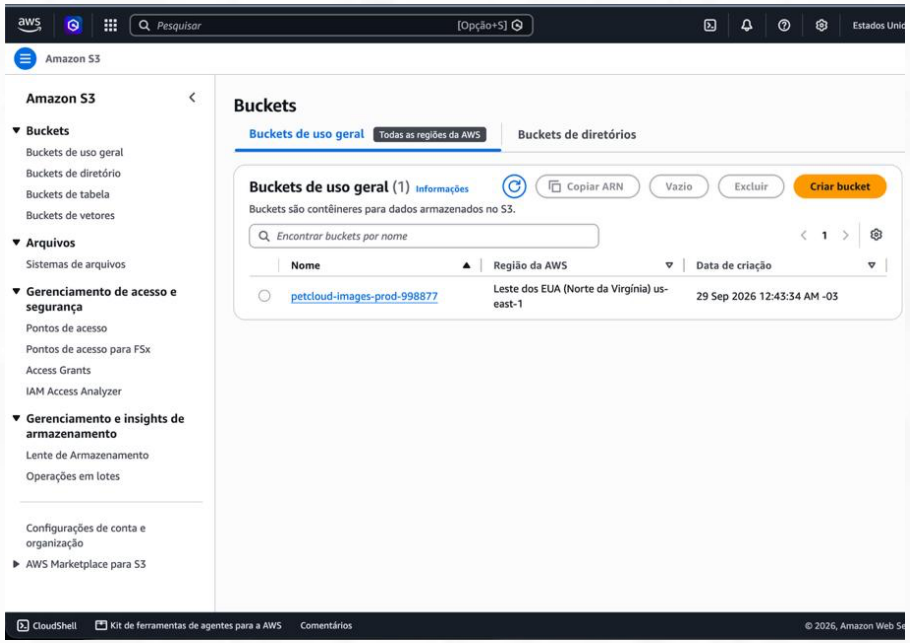
Marina finally picks up Bidu. PetCloud's cloud architecture worked invisibly behind the scenes to solve the fragmentation, but in real life, the impact was deeply human: bringing a family together.

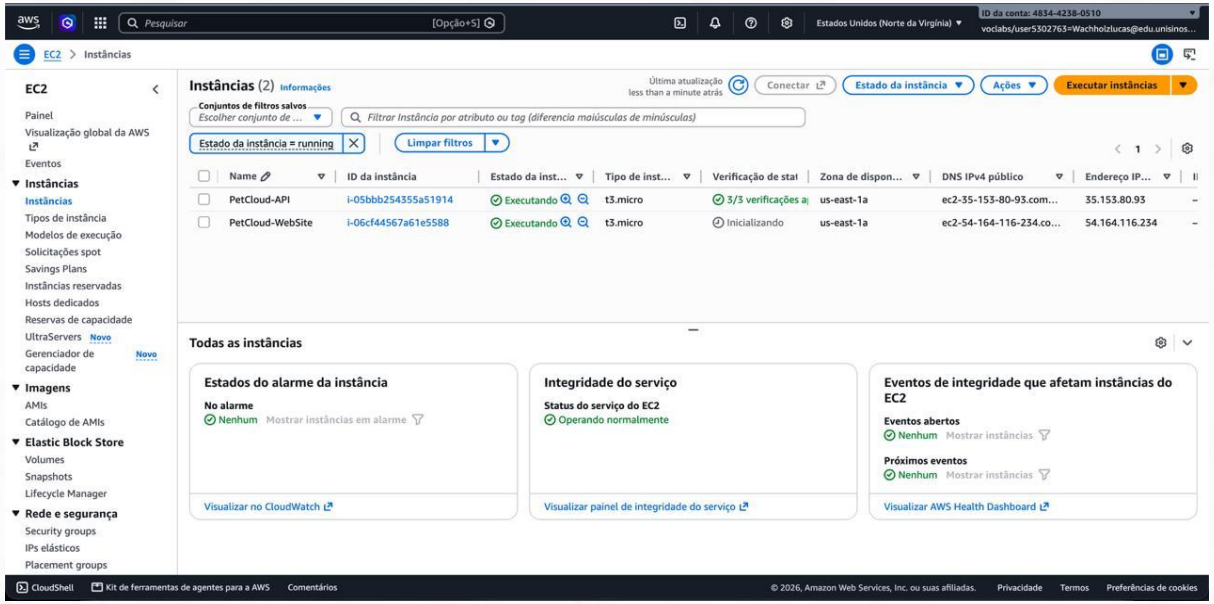
Deployment diagram

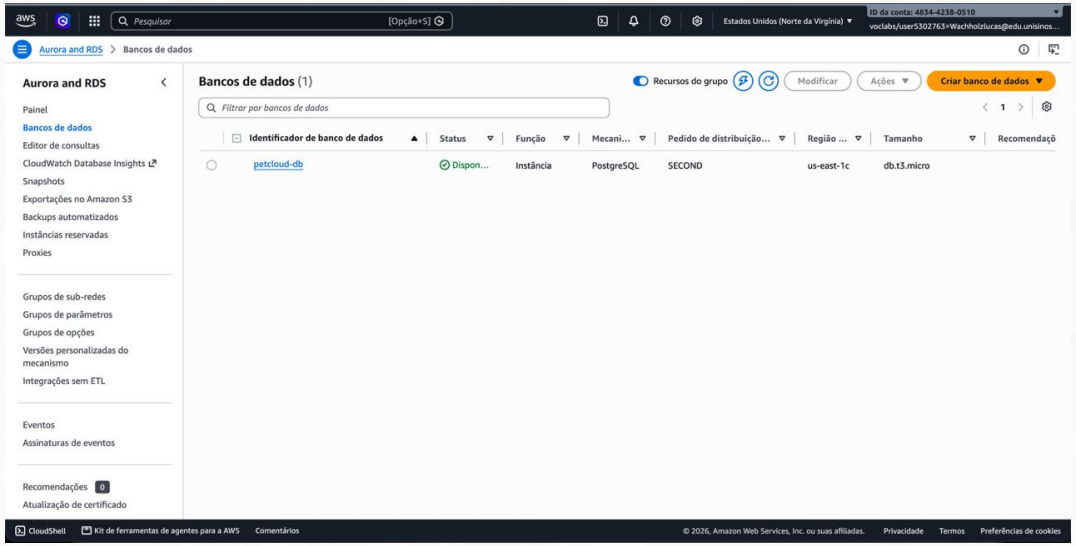


Demonstration of the Application









AWS CloudShell

Terminal: Used to create the instances.

Security: Configured the security and permissions to access.

Testing actual Website

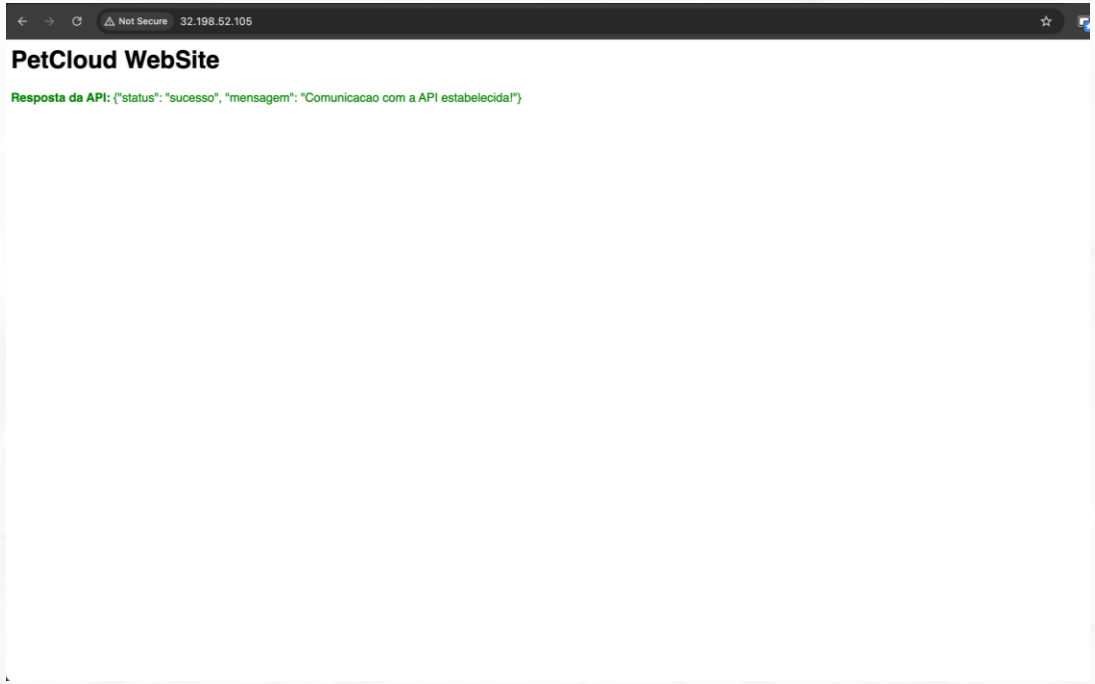
Connected website to the API, sending a default response and showing it on the screen.

```python
from http.server import BaseHTTPRequestHandler, HTTPServer
import json

class APIHandler(BaseHTTPRequestHandler):
    def do_GET(self):
        self.send_response(200)
        self.send_header('Content-type', 'application/json')
        self.end_headers()
        resposta = {"status": "sucesso", "mensagem": "Comunicacao com a API estabelecida!"}
        self.wfile.write(json.dumps(resposta).encode('utf-8'))

HTTPServer(('', 80), APIHandler).serve_forever()
```

api.py




```python
from http.server import BaseHTTPRequestHandler, HTTPServer
import urllib.request

class WebHandler(BaseHTTPRequestHandler):
    def do_GET(self):
        try:
            resposta = urllib.request.urlopen("http://172.31.45.96")
            dados = resposta.read().decode('utf-8')
            html = f"<html><head><meta charset='utf-8'></head><body style='font-family: sans-serif;'><h1>PetCloud \
            self.send_response(200)
        except Exception as e:
            html = f"<html><body style='font-family: sans-serif;'><h1>Erro de Comunicacao</h1><p style='color: red;
            self.send_response(500)

        self.send_header('Content-type', 'text/html')
        self.end_headers()
        self.wfile.write(html.encode('utf-8'))

HTTPServer(('', 80), WebHandler).serve_forever()
```

web.py

JUSTIFICATION

Why PetCloud Delivers Value



Focused Execution

Streamlined scope ensures robust delivery.



Scalable Backbone

AWS cloud guarantees high reliability.



Community Impact

Accelerating successful pet adoptions.

JUSTIFICATION

Why Use This Architecture?



Amazon EC2

COMPUTE LAYER

Backend Services: Handles business logic and API requests.

Scalable Capacity: Configurable resources for growth.



Amazon RDS

MANAGED DATABASE

Structured Records: Stores users, shelters, and pets.

Managed Operations: Delegated provisioning and backups.

Relational Integrity: Guarantees structured adoption data.

JUSTIFICATION

Why Use This Architecture?

Amazon S3

OBJECT STORAGE & WEB APP

Photo Hosting: Stores high-resolution pet imagery.

Decoupled Media: Preserves database speed and size.

High Durability: Secure object retention across regions.

CloudFront: Runs the PetCloud web platform.

Amazon CloudWatch

MONITORING & LOGS

Activity Tracking: Observes real-time application behavior.

Log Aggregation: Collects server and system logs.

Proactive Health: Monitored resources and alarms.

Next Steps of PetCloud



Code



Stack

FULL SOLUTION

Frontend: React Vite – TypeScript

Backend: FAST API – Python

Database: PostgreSQL

Cost: lightweight system to run in t2.micro

Agility: fast coding, docs and shipping

Note: Coding will focus on the adoption journey.

PetCloud

Turning a frustrating search into a **family's new beginning.**

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